Oracle Database Backup & Recovery Guide

RMAN Disaster Recovery Procedures & Scenario Execution Logs

Scenario A: Non-System Datafile Corruption / Deletion Recovery

Scenario: A disk error corrupted one of the non-system datafiles (e.g., USER_TEST tablespace / data file).

Tasks:

- Create a dummy table inside a test tablespace and insert sample records.
- Perform a full backup.
- Simulate corruption or manually delete the datafile corresponding to that tablespace from the OS while the DB is running.
- Attempt to query the dummy table to observe the error.
- Use your backup to restore and recover only the affected datafile/tablespace.

Deliverable: Log output of the error encountered when querying the table, followed by the RMAN/ Restore logs showing a successful recovery.

ANSWER

Step 1: Create a User_Test Tablespace, Dummy Table, and Insert Sample Records

Connect as SYS or a user with DBA privileges (`sysdba`):

`CREATE TABLESPACE USER_TEST DATAFILE '/u01/app/oracle/oradata/ORCL/user_test01.dbf' SIZE 50M AUTOEXTEND ON;`

```
SQL> CREATE TABLESPACE USER_TEST DATAFILE '/u01/app/oracle/oradata/ORCL/user_test01.dbf'
SIZE 50M AUTOEXTEND ON;

Tablespace created.

SQL> SELECT FILE_ID,TABLESPACE_NAME,FILE_NAME,STATUS FROM DBA_DATA_FILES ORDER BY FILE_ID;

   FILE_ID TABLESPACE_NAME FILE_NAME                                    STATUS
---------- --------------- -------------------------------------------- ---------
         1 SYSTEM          /u01/app/oracle/oradata/ORCL/system01.dbf    AVAILABLE
         2 UNDOTBS4        /u01/app/oracle/oradata/ORCL/undotbs04.dbf   AVAILABLE
         3 SYSAUX          /u01/app/oracle/oradata/ORCL/sysaux01.dbf    AVAILABLE
         4 USER_TEST       /u01/app/oracle/oradata/ORCL/user_test01.dbf AVAILABLE
         5 TEST_TB         /u01/app/oracle/oradata/ORCL/testtb01.dbf    AVAILABLE
         7 USERS           /u01/app/oracle/oradata/ORCL/users01.dbf     AVAILABLE

6 rows selected.
```

`CREATE TABLE hr.prod_catalog (item_id NUMBER PRIMARY KEY, item_name VARCHAR2(50), created_at TIMESTAMP DEFAULT SYSTIMESTAMP) TABLESPACE user_test;`

```
SQL> CREATE TABLE hr.prod_catalog (item_id NUMBER PRIMARY KEY, item_name VARCHAR2(50),
created_at TIMESTAMP DEFAULT SYSTIMESTAMP) TABLESPACE user_test;

Table created.

SQL> DESC HR.PROD_CATALOG;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 ITEM_ID                                   NOT NULL NUMBER
 ITEM_NAME                                          VARCHAR2(50)
 CREATED_AT                                         TIMESTAMP(6)
```

Insert the table records:

```
SQL>
SQL> INSERT INTO hr.prod_catalog (item_id, item_name) VALUES (101, 'Widget A');

1 row created.

SQL> INSERT INTO hr.prod_catalog (item_id, item_name) VALUES (102, 'Widget B');

1 row created.

SQL> commit;

Commit complete.

SQL> SELECT * FROM HR.PROD_CATALOG;

   ITEM_ID ITEM_NAME                      CREATED_AT
---------- ------------------------------ -----------------------------------
       101 Widget A                       14-AUG-26 10.19.30.441101 AM
       102 Widget B                       14-AUG-26 10.19.38.686906 AM
```

Verify datafile ID and status:

```
SQL> SELECT file_id, file_name, tablespace_name, status FROM dba_data_files WHERE
tablespace_name = 'USER_TEST';

   FILE_ID FILE_NAME                                   TABLESPACE_NAME STATUS
---------- ------------------------------------------- --------------- ---------
         4 /u01/app/oracle/oradata/ORCL/user_test01.dbf USER_TEST       AVAILABLE
```

Step 2: Perform a Full Database & Archive Log Backup

Run RMAN to take a full backup including the new tablespace:

`RMAN> BACKUP DATABASE PLUS ARCHIVELOG;`

```
[oracle@localhost ~]$ rman target/

Recovery Manager: Release 19.0.0.0.0 - Production on Fri Aug 14 10:22:44 2026
Version 19.3.0.0.0

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connected to target database: ORCL (DBID=1766032546)

RMAN> BACKUP DATABASE PLUS ARCHIVELOG;

Starting backup at 14-AUG-26
current log archived
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=71 device type=DISK
```

Step 3: Simulate Datafile Corruption / Deletion

Identify the file location and manually delete it from the OS while the database is still mounted/open.

```
[oracle@localhost ~]$
[oracle@localhost ~]$ cd /u01/app/oracle/oradata/ORCL/
[oracle@localhost ORCL]$ ll
total 3646468
-rw-r-----. 1 oracle oracle  10600448 Aug 14 10:26 control01.ctl
drwxrwx---. 2 oracle oracle         6 Jul 31 15:08 onlinelog
-rw-r-----. 1 oracle oracle 209715712 Aug 14 10:23 redo01.log
-rw-r-----. 1 oracle oracle 209715712 Aug 14 10:24 redo02.log
-rw-r-----. 1 oracle oracle 209715712 Aug 14 10:25 redo03.log
-rw-r-----. 1 oracle oracle 681582592 Aug 14 10:24 sysaux01.dbf
-rw-r-----. 1 oracle oracle 964698112 Aug 14 10:24 system01.dbf
-rw-rw----. 1 oracle oracle 524296192 Aug 13 15:22 temp_new01.dbf
-rw-rw----. 1 oracle oracle 1073750016 Aug 14 10:24 testtb01.dbf
-rw-rw----. 1 oracle oracle 314580992 Aug 14 10:24 undotbs04.dbf
-rw-r-----. 1 oracle oracle   5251072 Aug 14 10:24 users01.dbf
-rw-rw----. 1 oracle oracle  52436992 Aug 14 10:24 user_test01.dbf
[oracle@localhost ORCL]$
[oracle@localhost ORCL]$
[oracle@localhost ORCL]$ mv user_test01.dbf /u01
[oracle@localhost ORCL]$
[oracle@localhost ORCL]$ ls -ltr
total 3595260
drwxrwx---. 2 oracle oracle         6 Jul 31 15:08 onlinelog
-rw-rw----. 1 oracle oracle 524296192 Aug 13 15:22 temp_new01.dbf
-rw-r-----. 1 oracle oracle 209715712 Aug 14 10:23 redo01.log
-rw-r-----. 1 oracle oracle 209715712 Aug 14 10:24 redo02.log
-rw-r-----. 1 oracle oracle 964698112 Aug 14 10:24 system01.dbf
-rw-rw----. 1 oracle oracle 314580992 Aug 14 10:24 undotbs04.dbf
-rw-r-----. 1 oracle oracle 681582592 Aug 14 10:24 sysaux01.dbf
-rw-rw----. 1 oracle oracle 1073750016 Aug 14 10:24 testtb01.dbf
-rw-r-----. 1 oracle oracle   5251072 Aug 14 10:24 users01.dbf
-rw-r-----. 1 oracle oracle 209715712 Aug 14 10:26 redo03.log
-rw-r-----. 1 oracle oracle  10600448 Aug 14 10:27 control01.ctl
[oracle@localhost ORCL]$
```

Next, flush the buffer cache or restart the database instance so Oracle is forced to read from disk:

```
SQL> ALTER SYSTEM FLUSH BUFFER_CACHE;

System altered.
```

Step 4: Attempt to Query the Dummy Table

Execute a query against the dummy table to observe the physical I/O error:

```
SQL> SELECT * FROM hr.prod_catalog;
SELECT * FROM hr.prod_catalog
              *
ERROR at line 1:
ORA-01116: error in opening database file 4
ORA-01110: data file 4: '/u01/app/oracle/oradata/ORCL/user_test01.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

Step 5: Restore and Recover the Affected Datafile / Tablespace

Connect to RMAN and perform a Datafile / Tablespace Recovery without taking the entire database offline:

```
rman target /

-- Take only the damaged datafile offline (or tablespace)
RMAN> ALTER DATABASE DATAFILE 7 OFFLINE;

-- Restore the physical file from backup
RMAN> RESTORE DATAFILE 7;

-- Apply redo/archive logs to bring it up to current time
RMAN> RECOVER DATAFILE 7;

-- Bring the datafile back online
RMAN> ALTER DATABASE DATAFILE 7 ONLINE;
```

Step 6: Verification Post-Recovery

```
SQL> SELECT * FROM hr.prod_catalog;

   ITEM_ID ITEM_NAME                      CREATED_AT
---------- ------------------------------ -----------------------------------
       101 Widget A                       14-AUG-26 10.19.30.441101 AM
       102 Widget B                       14-AUG-26 10.19.38.686906 AM
```

Scenario B: Point-in-Time Recovery (PITR) after Unwanted TRUNCATE

Scenario: An application bug executed an unwanted DROP TABLE or DELETE statement at 10:30 AM. You need to recover the database state to 10:25 AM.

Tasks:

- Record the current time. Insert initial test data into a table.

- Perform a log backup/archive log backup.
- Record the exact timestamp, then simulate a disaster (e.g., TRUNCATE TABLE accounts;).
- Perform an Incomplete Recovery / Point-in-Time Recovery back to 1 minute before the truncation took place.
- Open the database with RESETLOGS.

Deliverable: A summary documenting the target timestamp/SCN used for recovery and proof that the dropped data was restored.

ANSWER

Step 1: Insert Initial Test Data & Note Timestamp

Connect to SQL*Plus as a user with write privileges (e.g., HR or SCOTT) and populate initial data:

```
SQL> SELECT * FROM HR.PROD_CATALOG;

   ITEM_ID ITEM_NAME                      CREATED_AT
---------- ------------------------------ ----------------------------------
       101 Widget A                       14-AUG-26 10.19.30.441101 AM
       102 Widget B                       14-AUG-26 10.19.38.686906 AM

SQL> SELECT CURRENT_TIMESTAMP FROM DUAL;

CURRENT_TIMESTAMP
---------------------------------------------------------------------------
14-AUG-26 04.06.31.282955 PM +05:30
```

Step 2: Take Full & Archive Log Backups

Take a baseline backup before simulating the disaster:

```
RMAN> backup database plus archivelog;


Starting backup at 14-AUG-26
current log archived
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=111 device type=DISK
channel ORA_DISK_1: starting archived log backup set
channel ORA_DISK_1: specifying archived log(s) in backup set
input archived log thread=1 sequence=8 RECID=29 STAMP=1241178851
input archived log thread=1 sequence=9 RECID=30 STAMP=1241256377
input archived log thread=1 sequence=10 RECID=31 STAMP=1241259814
input archived log thread=1 sequence=11 RECID=32 STAMP=1241259875
input archived log thread=1 sequence=12 RECID=33 STAMP=1241280645
channel ORA_DISK_1: starting piece 1 at 14-AUG-26
channel ORA_DISK_1: finished piece 1 at 14-AUG-26
piece handle=/u01/app/oracle/fast_recovery_area/ORCL/backupset/2026_08_14/
o1_mf_annnn_TAG20260814T161046_o7xw5gr5_.bkp tag=TAG20260814T161046
comment=NONE
```

Step 3: Record Safe Timestamp & Simulate Disaster

Insert additional records at a known time, record the safe recovery timestamp, and then execute the simulated disaster (e.g., `TRUNCATE TABLE prod_catalog;`).

```
SQL> insert into prod_catalog (item_id,item_name) values (103,'Widget C');

SQL> SELECT * FROM HR.PROD_CATALOG;

   ITEM_ID ITEM_NAME                      CREATED_AT
---------- ------------------------------ -----------------------------------
       103 Widget C                       14-AUG-26 04.18.51.733092 PM
       101 Widget A                       14-AUG-26 10.19.30.441101 AM
       102 Widget B                       14-AUG-26 10.19.38.686906 AM

SQL> SELECT TO_CHAR(SYSDATE, 'YYYY-MM-DD HH24:MI:SS') AS safe_time,
DBMS_FLASHBACK.GET_SYSTEM_CHANGE_NUMBER AS safe_scn FROM DUAL;

SAFE_TIME           SAFE_SCN
------------------- ----------
2026-08-14 16:21:01    3882061

SQL> /

SAFE_TIME           SAFE_SCN
------------------- ----------
2026-08-14 16:22:48    3882101

SQL> TRUNCATE TABLE HR.PROD_CATALOG;

Table truncated.

SQL> SELECT COUNT(*) FROM HR.PROD_CATALOG;

  COUNT(*)
----------
         0
```

Step 4: Perform Point-in-Time Recovery (PITR) via RMAN

Because Point-in-Time Recovery affects the entire database (incomplete recovery), the database must be shut down and started in `MOUNT` mode.

1. Backup Current Redo Logs & Shut Down

```
-- Archive active logs so the latest redo is available
SQL> ALTER SYSTEM SWITCH LOGFILE;
SQL> SHUTDOWN IMMEDIATE;
SQL> STARTUP MOUNT;
```

2. Execute RMAN Incomplete Recovery to Target Time

Launch RMAN and set the target time to 04:21:00 PM (before the TRUNCATE occurred):

```
RUN {
  -- Set target time format and target point
  SET UNTIL TIME "TO_DATE('2026-08-14 16:21:00', 'YYYY-MM-DD HH24:MI:SS')";
  -- Alternatively, recover by SCN:
  -- SET UNTIL SCN 3882061;
  RESTORE DATABASE;
  RECOVER DATABASE;
}
```

```
RMAN>
Starting restore at 14-AUG-26
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=46 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00001 to /u01/app/oracle/oradata/ORCL/system01.dbf
channel ORA_DISK_1: restoring datafile 00002 to /u01/app/oracle/oradata/ORCL/undotbs04.dbf
channel ORA_DISK_1: restoring datafile 00003 to /u01/app/oracle/oradata/ORCL/sysaux01.dbf
channel ORA_DISK_1: restoring datafile 00004 to /u01/app/oracle/oradata/ORCL/
user_test01.dbf
channel ORA_DISK_1: restoring datafile 00005 to /u01/app/oracle/oradata/ORCL/testtb01.dbf
channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/users01.dbf
channel ORA_DISK_1: reading from backup piece /u01/app/oracle/fast_recovery_area/ORCL/
backupset/2026_08_14/o1_mf_nnndf_TAG20260814T161051_o7xw5nf6_.bkp
channel ORA_DISK_1: piece handle=/u01/app/oracle/fast_recovery_area/ORCL/backupset/
2026_08_14/o1_mf_nnndf_TAG20260814T161051_o7xw5nf6_.bkp tag=TAG20260814T161051
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:01:05
Finished restore at 14-AUG-26

RMAN>
Starting recover at 14-AUG-26
using channel ORA_DISK_1

starting media recovery
media recovery complete, elapsed time: 00:00:01

Finished recover at 14-AUG-26
```

Step 5: Open Database with RESETLOGS

After incomplete recovery completes, open the database with `RESETLOGS` to reset log sequence numbers and establish a new timeline:

```
RMAN> ALTER DATABASE OPEN RESETLOGS;
```

Step 6: Verification Post-Recovery (Data Restored)

```
SQL> SELECT * FROM HR.PROD_CATALOG;

   ITEM_ID ITEM_NAME                      CREATED_AT
---------- ------------------------------ -----------------------------------
       103 Widget C                       14-AUG-26 04.18.51.733092 PM
       101 Widget A                       14-AUG-26 10.19.30.441101 AM
       102 Widget B                       14-AUG-26 10.19.38.686906 AM
```